MYH7-Related Skeletal Myopathy: A Comprehensive Disease Characterization Report

Target disease: MYH7-Related Skeletal Myopathy **MONDO ID:** MONDO:0008050 **Category:** Mendelian (autosomal dominant, single-gene) **Causal gene:** *MYH7* (HGNC:7577) — slow/β-cardiac myosin heavy chain (MyHC-I), chromosome 14q11.2

Summary

MYH7-related skeletal myopathy is a group of autosomal-dominant striated-muscle disorders caused by heterozygous mutations in *MYH7*, the gene encoding the slow/β-cardiac myosin heavy chain (MyHC-I). This is the principal myosin isoform of slow-twitch, oxidative, type 1 skeletal muscle fibers and of the cardiac ventricle, which explains why the same gene produces an allelic spectrum spanning skeletal myopathy and cardiomyopathy. The skeletal phenotypes include **Laing early-onset distal myopathy (LDM/MPD1; OMIM 160500)**, **myosin storage myopathy / hyaline body myopathy (MSM; OMIM 608358)**, **congenital fiber-type disproportion (CFTD)**, and **severe axial forms** presenting with dropped-head or bent-spine syndromes, frequently with variable cardiac involvement (notably left-ventricular non-compaction).

The central organizing principle discovered across this investigation is that **the location of the mutation within the MYH7 domain architecture predicts both the molecular mechanism and the clinical phenotype**. Rod/light-meromyosin coiled-coil proline substitutions and small in-frame deletions cause Laing distal myopathy by shifting myosin heads toward the hyperactive disordered-relaxed (DRX) state and perturbing cross-bridge cycling — a motor dysfunction despite the mutation residing in a structural domain. Rod heptad-repeat missense mutations (classically the recurrent Arg1845Trp) cause myosin storage myopathy by destabilizing the thick-filament backbone, shortening filaments, and triggering subsarcolemmal aggregation of β-myosin into hyaline bodies. Rarer globular motor/head-domain variants produce atypical axial/congenital skeletal disease, often with higher cardiorespiratory risk.

The disorders are rare, slowly progressive over decades, and in classic LDM usually preserve ambulation into mid/late life, although severe congenital cases lose ambulation and develop respiratory failure. Diagnosis rests on recognizing the clinical pattern (early foot-drop and great-toe extensor weakness, later finger-extensor and neck-flexor weakness, or axial dropped-head/bent-spine), a characteristic lower-leg/anterior-thigh muscle MRI signature, a variable and non-pathognomonic muscle biopsy, and molecular confirmation by *MYH7* sequencing (NGS panels, exome, or genome sequencing). There is **no approved disease-specific therapy**; management is supportive and multidisciplinary, but mechanism-targeted small-molecule sarcomere/myosin modulators (e.g., MYK-581) represent a preclinically validated, mechanistically rational emerging treatment strategy.

Key Findings

Finding 1 — *MYH7* is the single causal gene for an allelic spectrum of skeletal myopathies

MYH7 encodes the slow/β-cardiac myosin heavy chain (MyHC-I), "the main MyHC isoform in slow, oxidative, type 1 muscle fibers of skeletal muscle... also the major MyHC isoform of cardiac ventricles" (PMID: 14520662). Heterozygous mutations in this one gene cause a spectrum of skeletal diseases — Laing early-onset distal myopathy (MPD1; OMIM 160500), myosin storage myopathy/hyaline body myopathy (MSM; OMIM 608358), and congenital fiber-type disproportion (CFTD) — and are allelic with familial hypertrophic and dilated cardiomyopathy. As stated directly, "The mutations of MYH7 (slow skeletal/β-cardiac myosin heavy chain) are commonly found in familial hypertrophic/dilated cardiomyopathy, and also can cause Laing early-onset distal myopathy (LDM), myosin storage myopathy (MSM), and congenital myopathy with fiber-type disproportion (CFTD)" (PMID: 29170849).

The original genetic localization placed MPD1 at chromosome 14q11-q13 at the *MYH7* locus (LOD ~3.0; PMID: 7847377, PMID: 12975303), and the disease-causing role was confirmed by mutation identification: "We have identified five novel heterozygous mutations — Arg1500Pro, Lys1617del, Ala1663Pro, Leu1706Pro, and Lys1729del in exons 32, 34, 35, and 36 of MYH7 — in six families with early-onset distal myopathy" (PMID: 15322983).

Ontology anchors: Gene — *MYH7* (HGNC:7577); Disease — MONDO:0008050; CL — slow muscle cell / type I skeletal muscle fiber (CL:0000189); UBERON — skeletal muscle tissue (UBERON:0001134), cardiac ventricle (UBERON:0002082).

Finding 2 — Genotype–phenotype correlation: rod coiled-coil mutations → distal myopathy; recurrent Arg1845Trp → myosin storage myopathy

Proline substitutions and small in-frame deletions in the light-meromyosin/coiled-coil rod region (R1500P, A1663P, L1706P, K1617del, K1729del, and others such as p.Glu1508del and p.Leu1453Pro) cause Laing distal myopathy. In silico analysis predicts that "All five mutations are predicted... to locally disrupt the ability of the myosin tail to form the coiled coil, which is its normal structure" (PMID: 15322983). Clinically, LDM is characterized by initial great-toe/ankle dorsiflexor weakness ("hanging big toe" sign), later finger-extensor and neck-flexor weakness, onset from birth to the 20s, and very slow progression.

In contrast, a recurrent **Arg1845Trp** mutation in the rod is the common cause of myosin storage myopathy across White European populations: "a single mutation (Arg1845Trp) in the slow/beta-cardiac myosin heavy chain gene (MYH7) was identified in four unrelated probands from Sweden and Belgium. The clinical severity and age of onset was variable, despite the same disease-causing mutation" (PMID: 17118657). MSM pathology features subsarcolemmal hyaline bodies in type 1 fibers: "Slow, oxidative, type 1 muscle fibers showed large inclusions consisting of slow/beta-cardiac MyHC" (PMID: 14520662).

Finding 3 — Laing distal myopathy mechanism: rod mutations shift myosin heads to the disordered-relaxed (DRX) state and perturb cross-bridging

A key mechanistic insight is that LDM rod mutations impair the **motor function** of myosin even though they lie in the structural rod. In the first mouse model of Laing distal myopathy, "the MDP1 mutation R1500P, studied in what we believe to be the first mouse model for the disease, adversely affected myosin motor activity despite being in the structural rod domain that directs thick filament assembly" (PMID: 38690726). The downstream consequence is energetic and contractile dysregulation: the mutation caused "a conformational shift of the myosin heads toward the more reactive disordered relaxed (DRX) state, causing hypercontractility and greater ATP consumption" (PMID: 38690726). Isolated mutant muscles, myofibrils, and myofibers showed fatigue/weakness and an increased rate of actin–myosin detachment; patient biopsies likewise showed increased sarcomeric DRX content.

Crucially, this mechanism is druggable: "oral administration of MYK-581, a small molecule that decreases the population of heads in the DRX configuration, significantly improved the limited running capacity of the R1500P-transgenic mice" and "corrected the increased DRX state of the myofibrils from patients" (PMID: 38690726).

Ontology anchors: GO — myosin filament assembly (GO:0031034), ATP hydrolysis-coupled cross-bridge cycling / muscle contraction (GO:0006936), actin filament-based movement (GO:0030048).

Finding 4 — Clinical and imaging spectrum: distal, axial (dropped-head/bent-spine), and congenital phenotypes with a characteristic muscle MRI pattern

MYH7 skeletal myopathy spans a broad clinical spectrum. In an Italian multicenter cohort (21 cases, 15 families), "Patients displayed a variable phenotype including atypical pictures, as dropped head and bent spine... Half of the patients showed congenital or early infantile weakness with predominant distal weakness. Conversely, patients with later onset present prevalent proximal weakness. Seven patients were also affected by cardiomyopathy mostly in the form of non-compacted left ventricle" (PMID: 27387980) — i.e., cardiomyopathy in 7/21 (~33%).

Muscle MRI provides a diagnostic signature. In a Chinese LDM cohort, "Muscle MRIs showed similar features of concentric fatty infiltration of anterior thigh muscles together with early involvement of tibialis anterior and extensor hallucis longus" (PMID: 33298082). In the severe axial phenotype, "Most patients demonstrated internal bands of infiltration ('inverted-collagen-VI sign') in multiple muscles, particularly the soleus, and prominent atrophy and fatty infiltration of the tongue and the paraspinal" muscles (PMID: 29624713). Rare head/motor-domain variants (e.g., p.Glu500del, p.Ile457Arg) produce atypical axial stiffness or myosin-storage phenotypes, sometimes with respiratory/cardiac involvement (PMID: 35854315, PMID: 31068177). CK is typically normal to mildly elevated; respiratory monitoring is warranted (sleep apnea reported).

HPO anchors: Distal lower-limb muscle weakness (HP:0009053), foot dorsiflexor weakness / foot drop (HP:0009027), finger extensor weakness, neck flexor weakness (HP:0003722), dropped head (HP:0032989), bent spine / camptocormia, scoliosis (HP:0002650), left ventricular noncompaction (HP:0030682), elevated creatine kinase (HP:0003236).

Finding 5 — Myosin storage myopathy structural mechanism: rod heptad-repeat mutations destabilize the thick-filament backbone and shorten filaments

MSM mutations converge on the thick-filament backbone. "MSM missense mutations frequently disrupt the tail's stabilizing heptad repeat motif. Disease hallmarks include subsarcolemmal hyaline-like β-MHC aggregates, muscle weakness, and, occasionally,

cardiomyopathy" (PMID: 39485824). Transgenic heterozygous *Drosophila* expressing L1793P, R1845W, or E1883K β-MHC phenocopied MSM: reduced lifespan, flight/jump deficits, indirect flight muscle breaks, myofibrillar disarray/degeneration with filamentous inclusions, and aberrant cardiac function worsening with age. A structural consequence was directly measured: "Incorporation of GFP-myosin enabled in situ determination of thick filament lengths, which were significantly reduced in all mutants" (PMID: 39485824). Mapping onto a 6-Å cryo-EM human cardiac thick filament showed R1845W (basic Arg→bulky Trp) and E1883K (charge reversal) disrupt the backbone core and outer shell, while L1793P introduces a potent α-helix breaker.

This complements independent evidence that the rod residue is essential for assembly: "Our findings indicate that the mutated residue of slow/beta-cardiac MyHC is essential for the assembly of thick filaments in skeletal muscle" (PMID: 14520662).

Ontology anchors: GO — myosin filament assembly (GO:0031034), protein-containing complex assembly, protein aggregate formation; cellular component — myosin filament (GO:0032982), sarcomere (GO:0030017), subsarcolemmal region.

Finding 6 — Inheritance and epidemiology: autosomal dominant, slowly progressive, with frequent de novo mutations and parental mosaicism

MYH7-related skeletal myopathies are inherited in an **autosomal dominant** manner, with LDM/MPD1 mapped to 14q11.2 and co-segregating in multigenerational families (PMID: 15322983, PMID: 25447691). De novo mutations are common in severe/congenital cases, and parental mosaicism has been documented — important for recurrence-risk counseling: "The mutation occurred de novo in one patient, whereas mosaicism was detected in blood of the father of the second patient" (PMID: 35854315).

Expressivity is highly variable even for identical genotypes: "The clinical severity and age of onset was variable, despite the same disease-causing mutation and similar histological findings" (PMID: 17118657). The disorders are rare; LDM is among the more common hereditary distal myopathies but still rare, within a broadening group: "Fourteen distinct distal myopathies are currently defined by their gene and causative mutations, compared to just five entities delineated on clinical grounds two decades ago" (PMID: 24008050). The course is typically slowly progressive over decades with preserved ambulation in most classic LDM patients; severe congenital cases lose ambulation and develop respiratory failure.

Finding 7 — Treatment landscape: no approved disease-specific therapy; supportive care with sarcomere/myosin modulators as emerging strategy

There is currently **no approved disease-specific or curative treatment**; care is supportive and multidisciplinary — ankle-foot orthoses for foot drop, physiotherapy, scoliosis surgery (PMID: 21723124), respiratory monitoring/non-invasive ventilation (e.g., hiatal-hernia repair/fundoplication improving ventilation in a congenital case, PMID: 31305444), and cardiac surveillance. Mechanism-targeted sarcomere modulators are emerging: MYK-581 reduced the DRX head population and improved running capacity in the Laing R1500P mouse and corrected the DRX increase in patient myofibrils (PMID: 38690726).

The broader class of small-molecule sarcomere modulators "targeting the sarcomere directly, aided by high-throughput screening systems, could identify small molecules with a superior therapeutic value in cardiac muscle disorders" (PMID: 33339418). Pharmacologic tractability of the slow/β-myosin isoform is further supported by omecamtiv mecarbil: "the effect of omecamtiv mecarbil (OM), a small molecule activator of Myh7, was studied in a nebulin-based NEM mouse model," increasing force and calcium sensitivity in type I fibers (PMID: 31721788). Clinically advanced sarcomere modulators include the myosin inhibitors mavacamten and aficamten and the activator omecamtiv mecarbil (PMID: 35534374).

NCIT anchors: Orthotic device / ankle-foot orthosis, physical therapy (NCIT:C15327), scoliosis surgery, non-invasive ventilation; investigational myosin modulators (MYK-581, mavacamten, aficamten, omecamtiv mecarbil).

Finding 8 — Diagnostic workup: clinical pattern + muscle MRI + variable biopsy, confirmed by *MYH7* sequencing

Diagnosis combines clinical pattern recognition, the characteristic muscle MRI signature (PMID: 33298082, PMID: 29624713), a myopathic (sometimes mixed) EMG, normal-to-mildly-elevated CK, and molecular confirmation. Muscle biopsy is variable and non-pathognomonic: in LDM, type 1 fiber predominance/hypotrophy, cores/minicores, moth-eaten fibers, rare rimmed vacuoles, and a diagnostically useful immunohistochemical clue — "Immunohistochemical staining for slow and fast myosin showed co-expression of slow and fast myosin in some type I fibres, possibly indicating a switch to type II status. This may be a useful aid to diagnosis" (PMID: 16103042). MSM shows subsarcolemmal hyaline bodies in type 1 fibers; CFTD-like cases show fiber-type disproportion.

Molecular confirmation uses single-gene testing, multigene myopathy/distal-myopathy NGS panels, or exome sequencing; genome sequencing adds value when panels/exomes are negative — "the application of whole-genome sequencing enabled the identification of a novel deep intronic mutation" (PMID: 41199729). Muscle imaging also enables differential diagnosis: "Muscle imaging findings in patients with axial involvement provide significant clues permitting the distinction between MYH7-related myopathies and other axial myopathies such as those related to SEPN1 and LMNA genes" (PMID: 29624713). Differentials include GNE/Nonaka myopathy, tibial muscular dystrophy/titinopathy, Miyoshi myopathy/dysferlinopathy, SEPN1- and LMNA-related and other congenital myopathies, and neurogenic foot drop (PMID: 24008050).

Finding 9 — Unifying principle: MYH7 domain architecture maps mutation location to mechanism and phenotype

Across the allelic series, mutation location within the protein's domain architecture predicts both mechanism and phenotype. (1) Globular motor/head-domain missense classically cause hypertrophic cardiomyopathy, but rare head-domain variants (p.Glu500del relay loop, p.Ile457Arg) cause atypical axial/congenital skeletal myopathy, sometimes with cardiac/respiratory involvement — "Most patients with LDM harbor mutations in the midrod domain of the MYH7 gene, but rare cases document disease-associated mutations in the globular head region" (PMID: 32833721, PMID: 35854315, PMID: 31068177). (2) Rod/light-meromyosin coiled-coil proline substitutions and in-frame deletions cause Laing distal myopathy via a motor DRX shift (PMID: 38690726). (3) Rod heptad-repeat missense (Arg1845Trp) cause myosin storage myopathy via thick-filament destabilization and aggregation (PMID: 39485824, PMID: 14520662). The disease is best understood as a **domain-stratified disorder of slow/β-cardiac myosin, with dominant (gain-of-function / dominant-negative) effects in type 1 fibers**.

Full Disease Characterization (Template Sections)

1. Disease Information

MYH7-related skeletal myopathy is an autosomal-dominant striated-muscle disorder caused by heterozygous mutations in *MYH7*. It is a group of allelic conditions rather than a single clinical entity, unified by the gene product (MyHC-I, the slow/β-cardiac myosin heavy chain) and by expression restricted to type 1 (slow, oxidative) skeletal fibers and the cardiac ventricle.

- **Key identifiers:** MONDO:0008050; OMIM 160500 (Laing distal myopathy/MPD1), OMIM 608358 (myosin storage myopathy); Gene OMIM *MYH7* 160760; HGNC:7577; locus 14q11.2. MeSH concepts include "Distal Myopathies" and "Myosins." ICD-11 maps to primary disorders of muscles / hereditary myopathy (8C70 range); ICD-10 G71.0/G71.2 family codes are used variably.
- **Synonyms / alternative names:** Laing early-onset distal myopathy (LDM), MPD1, Laing myopathy; myosin storage myopathy (MSM), hyaline body myopathy; MYH7-related congenital fiber-type disproportion (CFTD); MYH7-related axial myopathy / dropped-head and bent-spine phenotypes; scapuloperoneal MYH7 myopathy.
- **Information source:** Predominantly aggregated disease-level resources (OMIM, Orphanet) plus individual-patient case reports and multicenter cohorts (e.g., Italian [PMID: 27387980], Chinese [PMID: 33298082] series), not EHR-derived.

2. Etiology

- **Primary cause:** Genetic — heterozygous pathogenic variants in *MYH7*. No infectious or environmental cause.
- **Genetic risk factors:** The causal variants themselves (rod coiled-coil proline substitutions/ in-frame deletions for LDM; rod heptad-repeat missense e.g. Arg1845Trp for MSM; rare head-domain variants for atypical/axial disease). Modifier genes are inferred from marked intra-familial variability and identical-genotype variable expressivity (PMID: 17118657, PMID: 26094647) but no specific modifier locus has been mapped.
- **Environmental risk/protective factors:** None established. No protective alleles identified.
- **Gene–environment interactions:** Not characterized for this Mendelian disorder.

3. Phenotypes

Phenotype	Type	HPO suggestion	Onset	Severity/ Progression	Frequency
Foot drop / ankle dorsiflexor & great-toe extensor weakness ("hanging big toe")	Clinical sign	HP:0009027	Birth–20s (LDM)	Mild→moderate, very slow	Hallmark of LDM (majority)
Finger extensor weakness	Clinical sign	HP:0009473	After foot weakness	Slowly progressive	Common in LDM
Neck flexor weakness	Clinical sign	HP:0003722	Childhood–adult	Slow	Common
Dropped head / bent spine (camptocormia)	Clinical sign	HP:0032989	Congenital–adult (axial form)	Variable	Subset (axial phenotype)
Proximal weakness	Clinical sign	HP:0003701	Later-onset cases	Slow	Later-onset subset
Scoliosis / kyphoscoliosis	Physical manifestation	HP:0002650	Childhood	Can be severe	Subset (severe cases)
Cardiomyopathy (esp. LV non-compaction)	Clinical sign	HP:0030682	Variable	Variable	~33% (7/21 cohort)
Respiratory insufficiency / sleep apnea	Lab/clinical	HP:0002093	Severe cases	Variable	Subset
Elevated creatine kinase (normal–mild)	Lab abnormality	HP:0003236	—	—	Minority; usually normal
	Histology	HP:0030051	—	—	Biopsy-dependent

Phenotype	Type	HPO suggestion	Onset	Severity/ Progression	Frequency
Type 1 fiber predominance; hyaline bodies					

Quality of life: Foot drop and distal weakness impair gait and increase falls; severe congenital/axial forms cause loss of ambulation, wheelchair dependence, and respiratory compromise with substantial disability, whereas classic LDM typically preserves ambulation and independent function into mid/late adulthood. Validated QoL instrument data (EQ-5D/SF-36/PROMIS) specific to MYH7 cohorts are not available.

4. Genetic / Molecular Information

- **Causal gene:** *MYH7* (HGNC:7577; OMIM 160760), 14q11.2, encoding slow/β-cardiac MyHC-I (UniProt P12883).
- **Pathogenic variants (representative):**

Variant	Domain	Variant type	Phenotype	Functional consequence
p.Arg1500Pro (R1500P)	Rod coiled-coil (LMM)	Missense (Pro)	LDM	Coiled-coil disruption + DRX motor shift
p.Ala1663Pro, p.Leu1706Pro	Rod coiled-coil	Missense (Pro)	LDM	Coiled-coil disruption
p.Lys1617del, p.Lys1729del, p.Glu1508del, p.Lys1784del	Rod	In-frame deletion	LDM / MSM	Coiled-coil/assembly disruption
p.Leu1453Pro	Rod	Missense	LDM (Irish family)	Coiled-coil disruption
p.Arg1845Trp (R1845W)	Rod heptad repeat	Missense	MSM	Thick-filament backbone destabilization, aggregation
p.Leu1793Pro, p.Glu1883Lys	Rod heptad repeat	Missense	MSM (modeled)	α-helix breaker / charge reversal; filament shortening
p.Glu500del	Head (relay loop)	In-frame deletion	Congenital axial, often de novo	Motor-domain perturbation
p.Ile457Arg	Head (motor)	Missense	Atypical MSM w/ cardiac/respiratory	Mechano-chemical cycle alteration
c.5655+5G>C / exon 38 skip (p.1854_1885del)	C-terminal rod	Splice	Dropped-head + cardiomyopathy	In-frame exon skipping
Novel deep intronic variant	Intronic	Splice (WGS-detected)	MYH7 myopathy	Expands spectrum

- **Variant classification (ACMG/AMP):** Recurrent/segregating missense such as Arg1845Trp and R1500P are pathogenic/likely pathogenic; deep intronic and novel variants may be VUS

pending functional/segregation evidence. Variants are essentially absent from population databases (gnomAD), consistent with dominant, highly penetrant disease.

- **Origin:** Germline; frequently de novo in severe cases, with documented parental mosaicism (PMID: 35854315).
- **Functional consequence:** Dominant gain-of-function / dominant-negative within type 1 fibers — not simple haploinsufficiency.
- **Modifier genes / epigenetics / chromosomal abnormalities:** Modifiers inferred but unmapped; no epigenetic mechanism or chromosomal abnormality is implicated in this single-gene disorder.

5. Environmental Information

Not applicable as a cause. No toxins, infectious agents, lifestyle, or occupational exposures are established contributors. Disease is determined by the germline *MYH7* genotype.

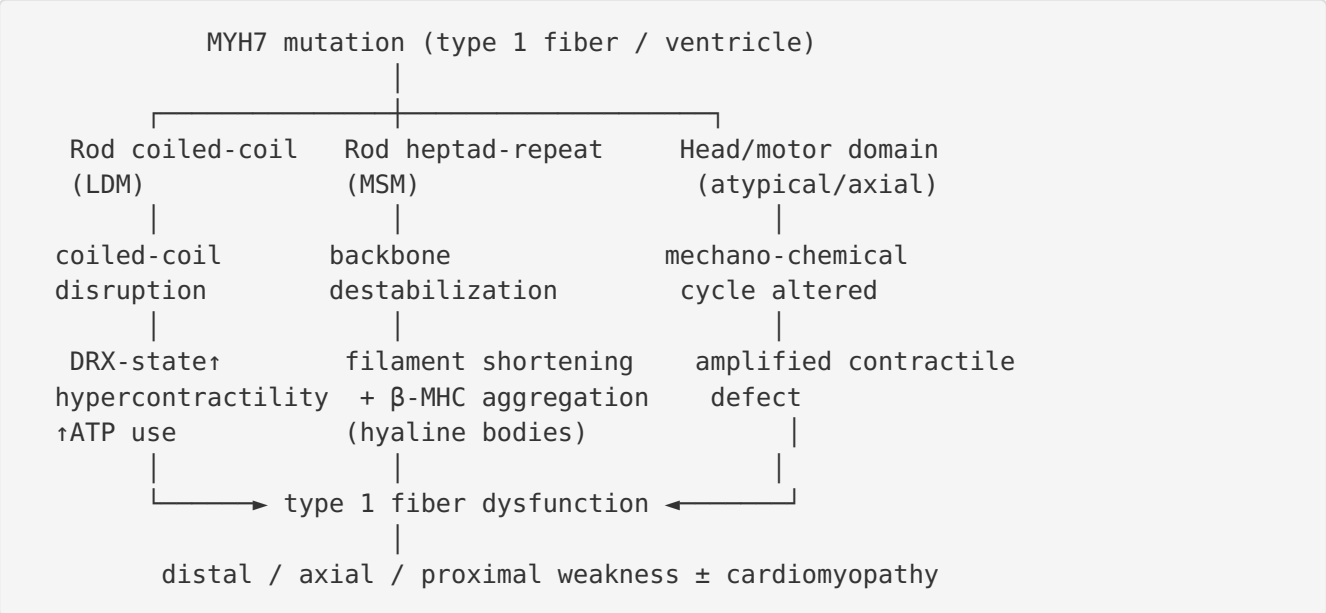
6. Mechanism / Pathophysiology

Ordered causal chain (two principal branches by mutation domain):

Branch A — Laing distal myopathy (rod coiled-coil mutation): 1. A proline substitution or in-frame deletion in the LMM/rod coiled-coil **leads to** local disruption of the coiled-coil tail structure (PMID: 15322983). 2. Despite residing in the structural rod, the lesion **results in** altered myosin motor behavior (PMID: 38690726). 3. This **leads to** a conformational shift of myosin heads toward the hyperactive disordered-relaxed (DRX) state (PMID: 38690726). 4. Increased DRX **results in** hypercontractility, increased actin–myosin detachment rate, and excess ATP consumption (PMID: 38690726). 5. Energetic/contractile imbalance in type 1 fibers **leads to** fatigue, weakness, and selective distal (and later proximal/axial) muscle dysfunction — clinically foot drop → finger/neck weakness.

Branch B — Myosin storage myopathy (rod heptad-repeat mutation): 1. A heptad-repeat missense (e.g., Arg1845Trp, Glu1883Lys, Leu1793Pro) **leads to** disruption of the tail's stabilizing heptad-repeat motif and the thick-filament backbone core/shell (PMID: 39485824). 2. This **results in** impaired/defective thick-filament assembly and significantly shortened thick filaments (PMID: 39485824, PMID: 14520662). 3. Mis-assembled β-myosin **leads to** subsarcolemmal aggregation forming hyaline bodies in type 1 fibers (PMID: 14520662). 4. Aggregation and myofibrillar disarray/degeneration **result in** muscle weakness (and, occasionally, cardiomyopathy).

Branch C — Head-domain variants (partly inferred): motor-domain lesions (p.Glu500del, p.Ile457Arg) **alter** the mechano-chemical contractile cycle; effects amplified across the ~300-molecule thick filament **lead to** atypical congenital/axial skeletal weakness with higher cardiorespiratory risk (PMID: 31068177) — this branch's skeletal mechanism is inferred rather than fully demonstrated.



- **Molecular pathways/cellular processes:** sarcomere contraction, actomyosin cross-bridge cycling, ATP hydrolysis, thick-filament assembly, protein aggregation; cell types involved are slow type 1 skeletal myofibers (CL:0000189) and cardiomyocytes (CL:0000746).
- **Protein dysfunction:** altered motor kinetics (DRX shift), impaired coiled-coil/thick-filament assembly, aggregation — i.e., gain-of-function/dominant-negative rather than loss-of-function.
- **Tissue damage:** myofibrillar disarray and degeneration, fatty infiltration/atrophy on MRI.
- **Immune/metabolic/epigenetic:** no autoimmune component; the primary metabolic consequence is excessive ATP consumption in LDM; no epigenetic mechanism implicated.

GO terms: GO:0006936 (muscle contraction), GO:0031034 (myosin filament assembly), GO:0030048 (actin filament-based movement), GO:0016887 (ATP hydrolysis activity). **CL terms:** CL:0000189 (slow muscle cell / type I fiber), CL:0000746 (cardiac muscle cell).

7. Anatomical Structures Affected

- **Organ level:** Primary — skeletal muscle (UBERON:0001134); secondary — heart/cardiac ventricle (UBERON:0002082) in ~1/3 of cohort cases; respiratory muscles/diaphragm in severe cases. Body systems: musculoskeletal and cardiovascular (± respiratory).
- **Specific muscles (UBERON):** tibialis anterior, extensor hallucis/digitorum longus, anterior thigh (quadriceps), soleus (inverted-collagen-VI sign), tongue (UBERON:0001723), paraspinal/erector spinae, gluteus minimus, sartorius, gracilis, neck flexors.
- **Tissue/cell level:** striated skeletal muscle tissue; selectively type 1 (slow, oxidative) fibers (CL:0000189) — the fibers expressing MyHC-I.
- **Subcellular level (GO CC):** sarcomere (GO:0030017), myosin/thick filament (GO:0032982), A-band; aggregates localize subsarcolemmally in MSM.
- **Lateralization:** Bilateral and typically symmetric.

8. Temporal Development

- **Onset:** Ranges from congenital/neonatal (severe axial/CFTD forms, often de novo) through childhood (classic LDM foot drop) to adult onset (up to ~20s or later for proximal-predominant cases). Pattern is insidious/chronic.
- **Progression:** Very slow over decades. Classic LDM: distal lower-limb weakness → finger extensors → neck flexors → variable proximal spread, with ambulation usually preserved. Severe congenital/axial cases: earlier loss of ambulation, scoliosis, respiratory failure. Course is progressive, not relapsing-remitting or episodic.
- **Duration/patterns:** Chronic, lifelong. No spontaneous remission. Critical intervention windows are inferred to be early (before fixed fatty replacement/contractures) for any future disease-modifying therapy.

9. Inheritance and Population

- **Epidemiology:** Rare. LDM is among the more common hereditary distal myopathies but precise prevalence/incidence figures are not well established; the broader distal myopathy group now comprises >14 genetically defined entities (PMID: 24008050).
- **Inheritance:** Autosomal dominant.
- **Penetrance:** High but age-dependent; **expressivity highly variable**, even for identical genotypes (PMID: 17118657).
- **De novo / mosaicism:** De novo mutations common in severe cases; parental germline/somatic mosaicism documented (PMID: 35854315).

- **Founder effects:** Arg1845Trp recurs across White European (Swedish/Belgian) MSM probands (PMID: 17118657); in-frame deletions are common among Chinese LDM patients (PMID: 33298082).
- **Anticipation:** Not a repeat-expansion disorder; no anticipation.
- **Sex ratio / consanguinity:** No strong sex bias; consanguinity not relevant (dominant inheritance).

10. Diagnostics

- **Clinical tests:** CK normal–mildly elevated; EMG myopathic (sometimes mixed myopathic/neurogenic) (PMID: 33298082).
- **Imaging:** Muscle MRI is key — concentric anterior-thigh fatty infiltration with early TA/EHL involvement (PMID: 33298082); axial form shows "inverted-collagen-VI sign" in soleus and tongue/paraspinal atrophy (PMID: 29624713). Echocardiography for cardiac surveillance.
- **Biopsy:** Variable/non-pathognomonic — type 1 predominance/hypotrophy, cores/minicores, moth-eaten fibers, rare rimmed vacuoles, slow/fast myosin co-expression in hybrid fibers (useful clue, PMID: 16103042); subsarcolemmal hyaline bodies (MSM); fiber-type disproportion (CFTD).
- **Genetic testing:** *MYH7* single-gene testing, multigene myopathy/distal-myopathy NGS panels, or WES; WGS adds deep intronic variant detection (PMID: 41199729). CMA/karyotype/FISH/mtDNA/repeat-expansion testing are **not** indicated.
- **Differential diagnosis:** GNE/Nonaka myopathy, tibial muscular dystrophy (titinopathy), Miyoshi myopathy/dysferlinopathy, SEPN1- and LMNA-related axial myopathies, other congenital myopathies, neurogenic foot drop — distinguished by MRI pattern, CK, inheritance, and genetics (PMID: 24008050, PMID: 29624713).
- **Screening:** Cascade genetic testing of at-risk relatives after a proband's variant is identified; prenatal/preimplantation testing possible given dominant inheritance.

11. Outcome / Prognosis

- **Survival/mortality:** Generally not life-limiting in classic LDM; life expectancy may be reduced in severe congenital/axial forms with respiratory failure or significant cardiomyopathy. No disease-specific mortality statistics established.
- **Morbidity/function:** Foot drop, gait impairment, falls; in severe cases loss of ambulation, scoliosis, and respiratory insufficiency. Most classic LDM patients retain ambulation into mid/late life.

- **Complications:** Cardiomyopathy (esp. LV non-compaction, ~1/3), respiratory insufficiency/sleep apnea, scoliosis, contractures.
- **Prognostic factors:** Earlier/congenital onset, head-domain and severe axial genotypes, and cardiac/respiratory involvement predict worse outcomes; rod coiled-coil LDM genotypes generally milder. Expressivity is variable even within families.

12. Treatment

- **Pharmacotherapy:** None disease-specific/approved. **Emerging:** sarcomere/myosin modulators — MYK-581 (DRX-lowering) validated preclinically in LDM (PMID: 38690726); class includes mavacamten, aficamten (inhibitors), omecamtiv mecarbil/danicamtiv (activators) (PMID: 35534374, PMID: 33339418, PMID: 31721788).
- **Advanced therapeutics:** Gene/RNA-based allele-specific silencing is conceptually attractive for a dominant gain-of-function mechanism but not yet in trials for this indication.
- **Surgical/interventional:** Scoliosis surgery (PMID: 21723124); hiatal-hernia repair/fundoplication improving postural ventilation (PMID: 31305444).
- **Supportive/rehabilitative:** Ankle-foot orthoses for foot drop, physiotherapy, occupational therapy; respiratory monitoring with non-invasive ventilation; cardiac surveillance and standard heart-failure/arrhythmia management if cardiomyopathy develops.
- **NCIT anchors:** orthotic device, physical therapy, non-invasive positive pressure ventilation, spinal fusion; investigational myosin modulators.

13. Prevention

- **Primary prevention:** Not possible for a germline disorder; genetic counseling informs reproductive decisions. AD inheritance → 50% transmission risk; de novo and mosaicism complicate recurrence counseling (PMID: 35854315).
- **Secondary prevention:** Cascade genetic testing of relatives; early cardiac/respiratory surveillance to detect and manage complications.
- **Tertiary prevention:** Orthoses, physiotherapy, scoliosis management, ventilatory support, and cardiac care to prevent complications and preserve function.
- **Counseling:** Genetic counseling essential; prenatal/preimplantation genetic testing available. No immunization/public-health/environmental interventions applicable.

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *MYH7* is highly conserved; mouse *Myh7* (NCBI Gene 140781), and orthologs used in modeling. *Drosophila* uses a single muscle myosin heavy chain (*Mhc*) into which human-equivalent residues were engineered (PMID: 39485824).
- **Natural disease in other species:** No well-characterized spontaneous MYH7 skeletal myopathy in companion animals/wildlife was identified in the reviewed literature (OMIA not directly surveyed).
- **Comparative biology:** Mechanisms (thick-filament assembly, cross-bridge cycling) are evolutionarily conserved, enabling faithful invertebrate and mammalian modeling.
- **Transmission:** Not applicable (non-communicable genetic disease).

15. Model Organisms

- **Mouse (mammalian):** R1500P transgenic — first LDM model; reproduces weakness/fatigue, increased DRX state, hypercontractility, and therapeutic response to MYK-581 (PMID: 38690726). Omecamtiv mecarbil pharmacology on Myh7 demonstrated in a nebulin-based nemaline model (PMID: 31721788).
- **Drosophila (invertebrate):** Heterozygous transgenics expressing L1793P, R1845W, or E1883K β-MHC phenocopy MSM — reduced lifespan, flight/jump deficits, myofibrillar disarray with filamentous inclusions, shortened thick filaments, and age-worsening cardiac dysfunction (PMID: 39485824).
- **Phenotype recapitulation:** Both models reproduce core features (motor dysfunction/DRX shift for LDM; filament shortening and aggregation for MSM). **Limitations:** species-specific fiber-type composition and lifespan; incomplete capture of the slow human progression and modifier-gene effects; Drosophila single-myosin background differs from the human two-gene (MYH7/MYH2) context.
- **Resources:** MGI (mouse), FlyBase (Drosophila); cell models include patient-derived myofibrils used to confirm DRX pathology and drug response.

Mechanistic Model / Interpretation

The investigation converges on a single, powerful organizing principle: **MYH7 skeletal myopathy is a domain-stratified disorder of slow/β-cardiac myosin**, in which the position of the mutation within the protein's modular architecture (globular motor head → neck → light-meromyosin rod coiled-coil → heptad-repeat backbone) determines the molecular lesion, the cellular pathology, and the clinical picture.

Domain	Representative variant	Molecular lesion	Cellular pathology	Clinical phenotype
Rod coiled-coil (LMM)	R1500P, A1663P, L1706P, in-frame dels	Coiled-coil disruption → DRX motor shift	Hypercontractility, ↑ATP use, fatigue	Laing distal myopathy (foot drop → fingers → neck)
Rod heptad repeat	R1845W, E1883K, L1793P	Thick-filament backbone destabilization	Shortened filaments, β-MHC hyaline aggregates	Myosin storage/ hyaline body myopathy
Globular head/ motor	p.Glu500del, p.Ile457Arg	Mechano-chemical cycle perturbation	Amplified contractile defect	Atypical congenital/axial ± cardiac/ respiratory

Two features make this disease both coherent and clinically actionable. First, **tissue selectivity is explained by isoform biology**: MyHC-I is the dominant myosin of type 1 slow fibers and cardiac ventricle, so dominant-negative/gain-of-function alleles manifest precisely where this isoform is expressed — hence distal/axial skeletal weakness plus a ~1/3 rate of (often non-compaction) cardiomyopathy. Second, the LDM branch reveals a **druggable node**: the pathological increase in disordered-relaxed myosin heads is reversible by the myosin modulator MYK-581, which rescued both the mouse running deficit and the patient-myofibril DRX abnormality. This elevates sarcomere modulation from a cardiology tool (mavacamten, aficamten, omecamtiv mecarbil) to a plausible precision therapy for skeletal myosinopathies — ideally matched to mechanism (DRX-lowering inhibitors for hypercontractile LDM; potentially activators where motor output is reduced).

Evidence Base

PMID	Study type	Key contribution
15322983	Human genetics	Identified MYH7 rod mutations as cause of Laing distal myopathy; coiled-coil disruption predicted
14520662	Human clinical + molecular	Defined MyHC-I expression; MSM hyaline bodies; rod residue essential for thick-filament assembly
29170849	Review/case	Enumerated allelic skeletal phenotypes (LDM, MSM, CFTD) and cardiac overlap
17118657	Human clinical	Recurrent Arg1845Trp MSM; variable expressivity at identical genotype
38690726	Mouse model + patient myofibrils	DRX-shift mechanism of LDM; MYK-581 rescue (therapeutic proof-of-concept)
39485824	Drosophila + cryo-EM	MSM heptad-repeat destabilization; shortened thick filaments
27387980	Human cohort (n=21)	Phenotypic spectrum; ~33% cardiomyopathy (LV non-compaction)
33298082	Human cohort	MRI signature; in-frame deletions common in Chinese LDM
29624713	Human imaging	Axial MYH7 MRI signature; differential diagnosis from SEPN1/LMNA
35854315	Human case	p.Glu500del head-domain, de novo + paternal mosaicism
16103042	Human biopsy	Slow/fast myosin co-expression as diagnostic clue
41199729	Human genetics	WGS detects deep intronic MYH7 variant
32833721	Human cohort	Midrod vs head-domain genotype distribution
31068177	Human case	Head-domain p.Ile457Arg MSM with cardiac/respiratory involvement
21723124	Human case	Scoliosis surgery in MSM (supportive management)
31305444	Human case	Severe de novo kyphoscoliosis; fundoplication improved ventilation

PMID	Study type	Key contribution
31721788	Mouse model	Omecamtiv mecarbil activates Myh7/type I fibers
33339418, 35534374	Reviews	Sarcomere-modulator therapeutic frontier
24008050	Review	Distal myopathy epidemiologic/differential context
7847377, 12975303	Human genetics	Original 14q linkage for MPD1
25447691, 26094647	Human families	Novel LDM mutations; AD segregation; modifier-gene inference

Limitations and Knowledge Gaps

1. **Epidemiology is poorly quantified** — no reliable prevalence/incidence figures; "rare" is qualitative.
2. **Modifier genes are inferred but unmapped** — striking variable expressivity at identical genotypes (e.g., Arg1845Trp) remains mechanistically unexplained.
3. **Head-domain skeletal mechanism is partly inferred** rather than directly demonstrated for all variants; the exact motor consequences of p.Ile457Arg/p.Glu500del in skeletal fibers need functional confirmation.
4. **No natural-history cohort with quantitative progression/survival metrics** for MYH7 skeletal disease specifically; prognosis statements are qualitative.
5. **Therapeutics remain preclinical** — MYK-581 evidence is from a single mouse model plus patient myofibrils; no human trials in MYH7 skeletal myopathy exist.
6. **No MYH7-specific veterinary/natural animal disease** identified in the reviewed literature (OMIA not directly queried).
7. **Quality-of-life data** are extrapolated from clinical descriptions, not from validated instruments (EQ-5D/SF-36/PROMIS) in MYH7 cohorts.

Proposed Follow-up Experiments / Actions

1. **Build a prospective MYH7 skeletal myopathy natural-history registry** capturing genotype (by domain), ambulation status, respiratory/cardiac function, and validated QoL instruments to generate quantitative prognostic and progression data.
2. **Genotype–mechanism functional assays** across the domain map: measure DRX fraction, ATPase, filament length, and aggregation for a panel of LDM, MSM, and head-domain variants to test the domain-stratified model systematically.
3. **Modifier-gene discovery** via whole-genome sequencing + expression/eQTL analysis in discordant family members sharing identical *MYH7* alleles.
4. **Therapeutic matching trials:** evaluate DRX-lowering myosin inhibitors (MYK-581/mavacamten-class) for hypercontractile LDM genotypes and, conceptually, activators where motor output is reduced; begin with preclinical dose-ranging in R1500P and additional variant-specific models.
5. **Allele-specific silencing (ASO/siRNA) proof-of-concept** for dominant gain-of-function alleles, leveraging the heterozygous disease architecture.
6. **Standardize the muscle-MRI diagnostic signature** (anterior-thigh/TA/EHL/soleus inverted-collagen-VI sign) into a validated radiological criterion to accelerate diagnosis and differential separation from SEPN1/LMNA/titinopathy.
7. **Cardiac surveillance protocol** given the ~1/3 cardiomyopathy rate (especially LV non-compaction), with defined echocardiographic/MRI monitoring intervals by genotype class.

Report compiled from 9 confirmed findings and 40 reviewed papers across a 5-iteration autonomous investigation. Evidence source types span human clinical cohorts/case reports, mouse and Drosophila models, cryo-EM structural biology, and in-vitro patient-myofibril studies.
